

Report Reflection: Lending Club Capstone

1. I dropped `interest_rate` as a leakage feature because Lending Club sets the interest rate based on its own internal risk judgement, so it already contains information about the target. If I had kept it, the model would likely look very accurate but it would be cheating by using Lending Club's decision after the fact instead of learning from application time borrower characteristics.
2. In my notebook, the neural network validation ROC-AUC was only slightly above logistic regression results, not dramatically better. This suggests neural networks are most worth the extra complexity when the data contain stronger nonlinear patterns or more complex structure than a simpler model can capture. For a structured tabular dataset like this one, a simpler model can already do much of the job
3. The threshold should shift lower, because that makes the model more likely to flag potentially risky borrowers instead of missing them. In my notebook, lowering the threshold improved recall substantially for the recommended Gradient Boosting model, which makes sense when false negatives are much more costly than false positives.
4. I scaled features after the train/test split so the scaler would learn only from the training data, not from the held out data. If I had scaled it first, the test set mean and standard deviation would leak into the training pipeline, which would make the final test results look better than they really are and weaken the validity of the evaluation.
5. For logistic regression, ordinal encoding a category like loan grade as 1-7 would force the model to treat the categories as equally spaced numeric steps, which may create a false linear relationship that does not actually exist. It matters less for decision trees because trees split on thresholds instead of assuming one smooth linear effect, but it can still impose an artificial ordering that is not ideal when the categories do not truly behave like measured numbers.